BAN 674 Homework Assignment 1 Report

Introduction

This report summarizes the data preparation, exploratory analysis, and visualizations completed in the Python notebook for Homework Assignment 1. The assignment required analysis of the ToyotaCorolla dataset and an initial exploration of the Market Street Wine dataset. The Toyota analysis included dataset summarization, mileage normalization, creation of fuel-type dummy variables, and partitioning into training, validation, and test sets. The wine analysis required a brief summary of the training dataset in preparation for further exploration.

1. Toyota Corolla Dataset

Dataset loading and initial inspection

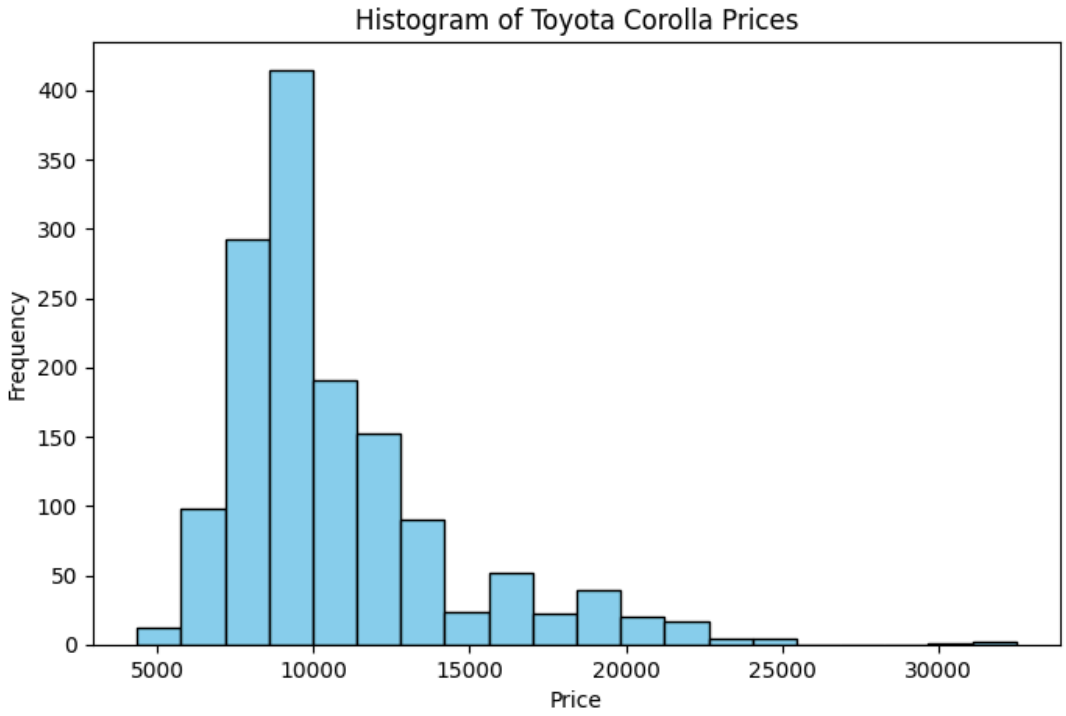
The notebook imported the required Python libraries and loaded `ToyotaCorolla.csv` into a pandas DataFrame named `toyota_df`. Column names were cleaned by replacing spaces with underscores.

The Toyota dataset contains 1,436 vehicles and 39 variables. The variables describe vehicle price, age, mileage, manufacturing information, fuel type, engine characteristics, and optional features such as air conditioning, airbags, ABS, power steering, and parking assistance.

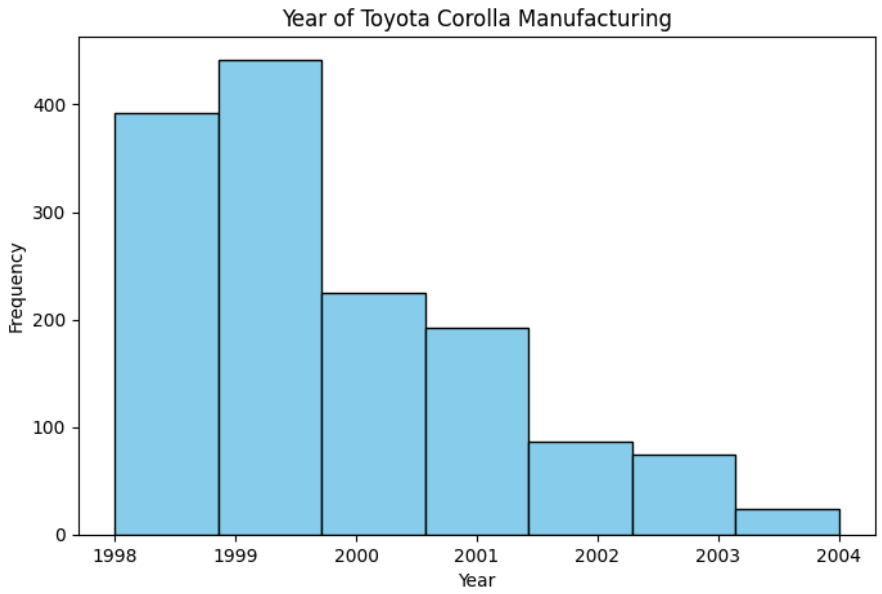
The notebook displayed the first and last five rows, reviewed the data types, and checked for missing values. No missing values were found in the dataset.

Dataset summary and observations

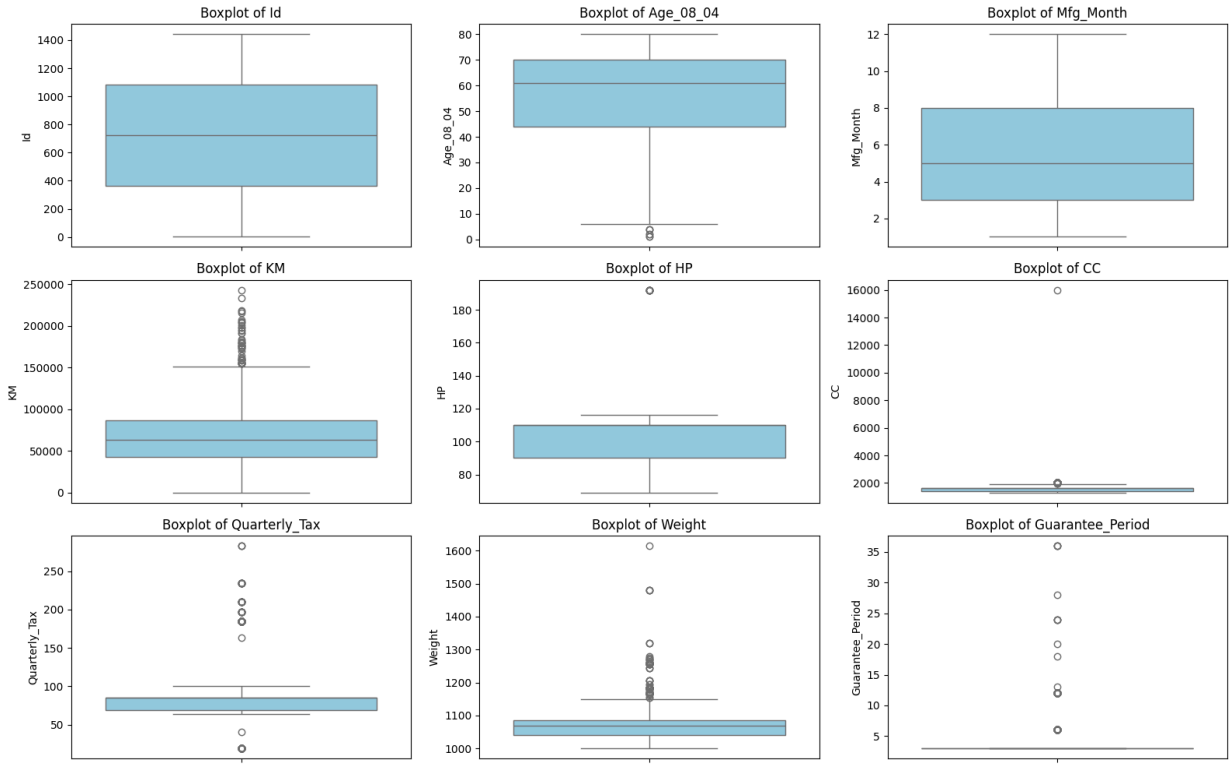
The average vehicle price was approximately 10,730.82, while the median price was 9,900. Prices ranged from 4,350 to 32,500. The price histogram showed a right-skewed distribution: most vehicles were priced between approximately 7,000 and 13,000, while relatively few vehicles had prices above 15,000.



The average vehicle age was approximately 56 months, with values ranging from 1 to 80 months. The manufacturing-year histogram showed that most vehicles were manufactured between 1998 and 2001. The number of vehicles declined for the later years, particularly 2003 and 2004.



The average horsepower was approximately 101.5, and most vehicles were concentrated near the 90-to-110 horsepower range. The boxplots also identified potential outliers in horsepower, engine displacement, quarterly tax, weight, mileage, and guarantee period. One especially unusual observation was an engine displacement value of 16,000 cc, which should be checked for possible data-entry or measurement issues.

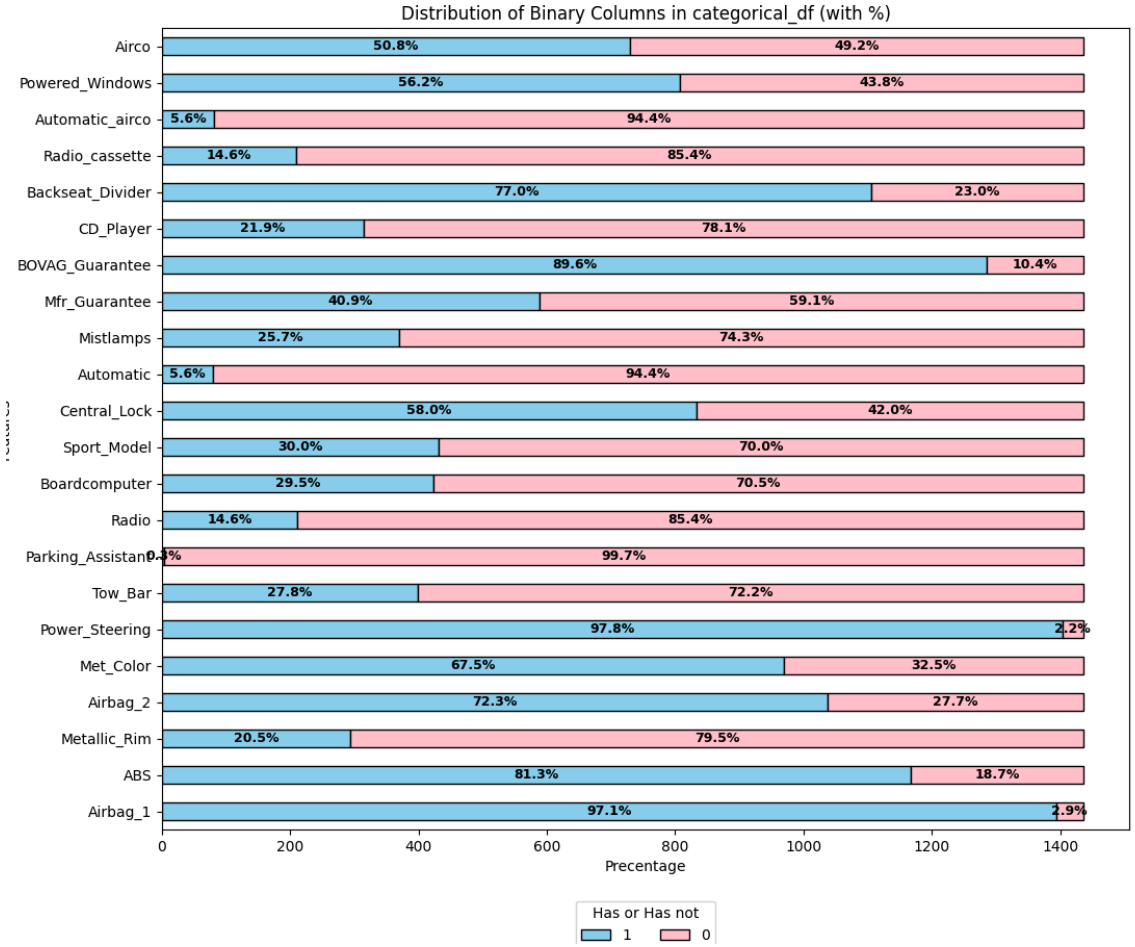


Categorical variables

The notebook identified binary variables containing only 0 and 1. These included transmission type, ABS, airbags, air conditioning, power steering, central locking, CD player, sport model, tow bar, and other equipment indicators.

I believe that the numbers 1 and 0 were assigned wrong. Normally 1 means that there is a feature whereas 0 means the feature is not included. It would seem like there would be more cars with automatic transmission than not as well as more cars with a radio then not.

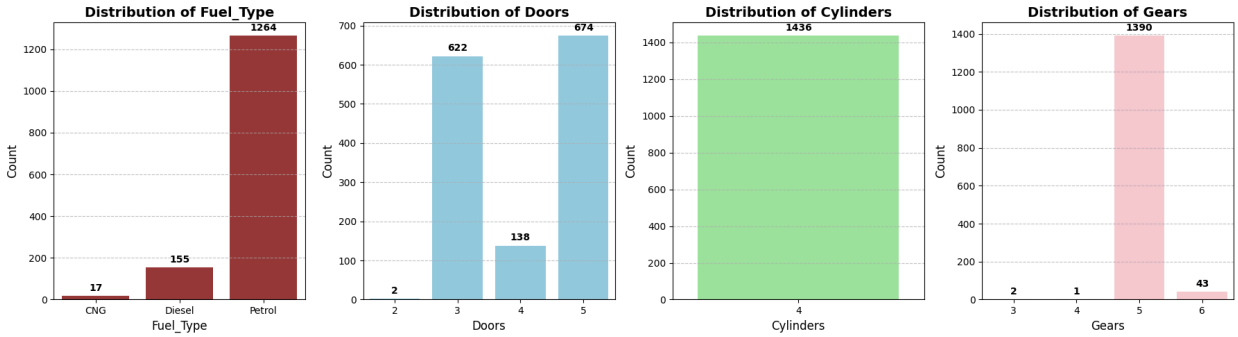
The binary-feature visualization showed that power steering and Airbag 1 were present in nearly all vehicles. ABS was present in 81.3% of vehicles, while air conditioning was present in approximately half of the observations. Automatic transmission appeared in only 5.6% of vehicles. The notebook identified this as a potentially surprising result that may warrant checking for possible data-labeling issues.



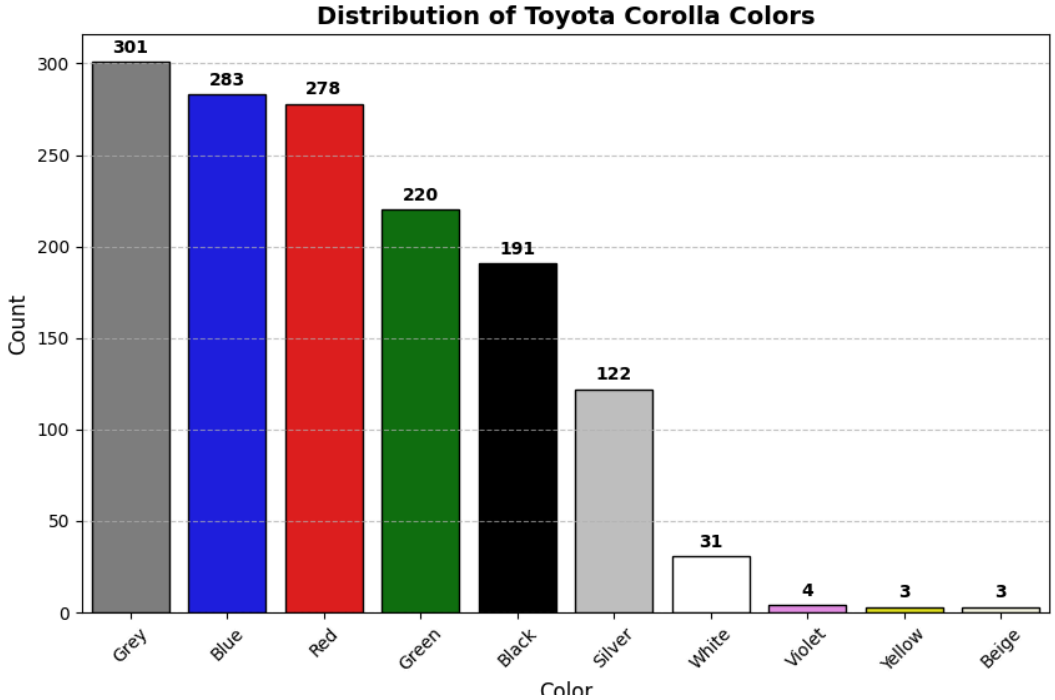
The dataset contained 319 unique Toyota Corolla model descriptions. Fuel type was strongly concentrated in petrol vehicles:

- Petrol: 1,264 vehicles, or 88.0%
- Diesel: 155 vehicles, or 10.8%
- CNG: 17 vehicles, or 1.2%

The most common body configurations had either five doors or three doors. All 1,436 vehicles were recorded as having four cylinders. Five-speed transmissions were also dominant, appearing in 1,390 vehicles, or 96.8% of the dataset. All Corollas are 4 cylinders



The color visualization showed that grey, blue, and red were the most common vehicle colors. Grey was the most frequent color with 301 vehicles, followed by blue with 283 and red with 278.



2. Toyota Data Preparation

Mileage normalization

The assignment required normalization of the kilometers variable. The notebook created a standardized version of `KM`called `normalized_KM` using the z-score formula:

$$Normalized\ KM\ = \frac{KM - mean(KM)}{standard\ deviation(KM)}$$

This transformation places mileage values on a common scale centered around zero, which is useful for many machine-learning methods.

Fuel-type dummy variables

The categorical `Fuel_Type` variable was converted into three dummy variables:

- `Fuel_CNG`
- `Fuel_Diesel`
- `Fuel_Petrol`

Each row receives a value of 1 for its fuel category and 0 for the other categories. These variables make the categorical fuel information usable in later statistical or machine-learning models.

Data partitioning

The Toyota data was randomly partitioned into three subsets using a fixed random state of 1:

- Training set: 718 observations, or 50%
- Validation set: 431 observations, or 30%
- Test set: 287 observations, or 20%

This partitioning satisfies the assignment requirement and creates separate datasets for model training, model selection, and final evaluation.

3. Market Street Wine Dataset

Dataset loading and initial summary

The notebook loaded `Wine.csv` into a DataFrame named `wine_df` and reviewed the first several rows, column names, data types, and descriptive statistics.

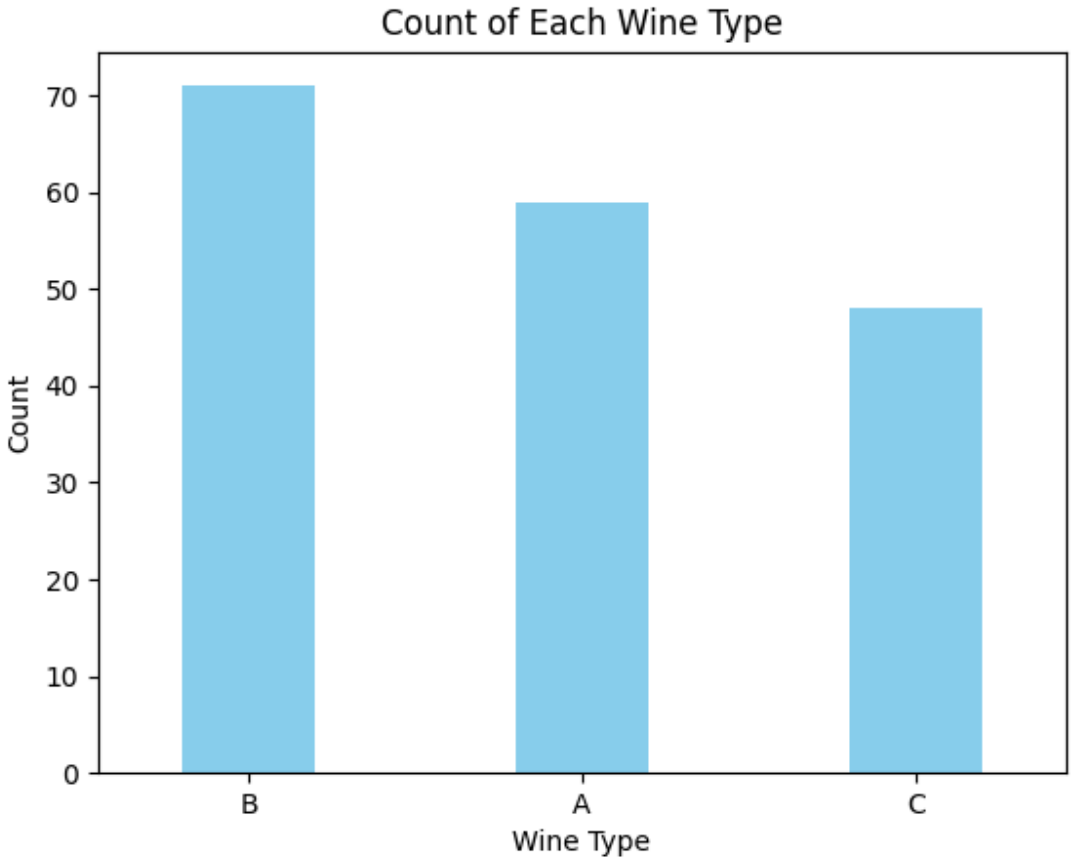
The dataset contains 178 observations and 14 original variables. It includes one categorical wine-type variable and thirteen chemical measurements, including alcohol, malic acid, ash, magnesium, phenols, flavanoids, color intensity, hue, and proline.

No missing values were reported. The average alcohol content was approximately 13.00, the average magnesium level was approximately 99.74, and the average proline level was approximately 746.89.

The wine-type bar chart showed three categories:

- Type B: 71 observations
- Type A: 59 observations
- Type C: 48 observations

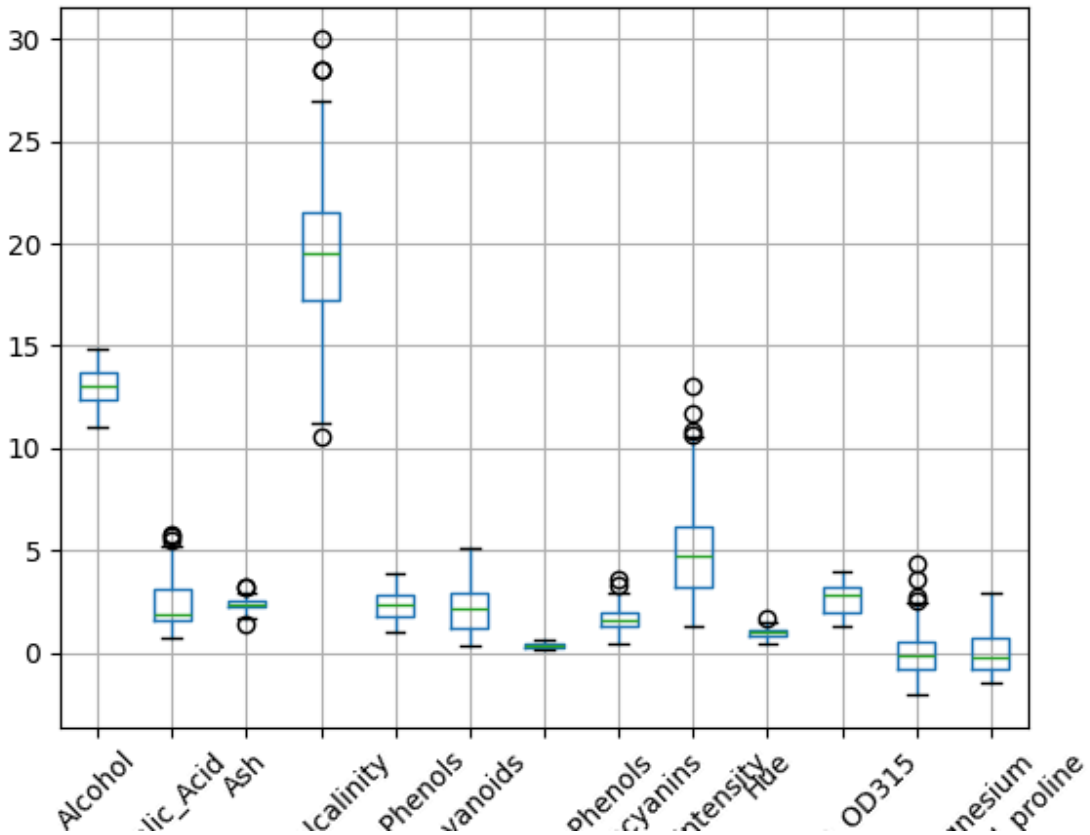
Type B was the largest group, while Type C was the smallest.



Wine preprocessing and visualization

The notebook standardized the `Magnesium` and `Proline` variables, creating `normalized_magnesium` and `normalized_proline`. A separate DataFrame was then created with the original magnesium and proline columns removed.

The wine boxplot displayed the distributions of the chemical variables. It showed variation across the measurements and several potential outliers, particularly in malic acid, ash alkalinity, proanthocyanins, color intensity, and proline. The normalized variables were centered closer to zero, demonstrating the effect of standardization.



The notebook provides an initial summary of the wine data; however, it does not explicitly label or verify that `Wine.csv` is the training-only file referenced in the assignment. It also does not document a separate train/test split for the wine data. Further case analysis is therefore left for the next stage of the assignment.

Conclusion

The notebook completed the main Toyota Corolla preparation requirements from Homework Assignment 1. It summarized the dataset using descriptive statistics and visualizations, normalized the kilometers variable, created fuel-type dummy variables, and partitioned the data into training, validation, and test sets.

The notebook also completed an initial exploration of the Market Street Wine dataset by describing its structure, reviewing descriptive statistics, visualizing wine-type counts and chemical-variable distributions, and normalizing magnesium and proline.

Overall, the analysis identified important patterns in both datasets. Toyota vehicles were predominantly petrol-powered, four-cylinder, five-speed vehicles, with prices concentrated in the lower-to-middle range and substantial mileage variation. The wine data contained three classes with different chemical measurements and several potential outliers that may be relevant in future modeling.